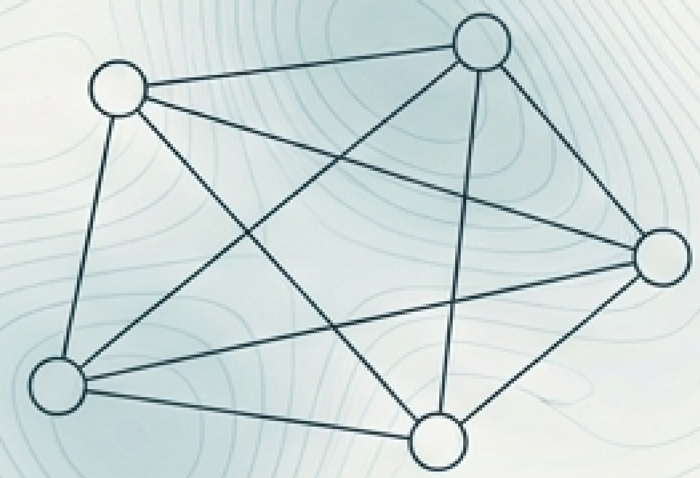
Portfolio Studio Architecture: Engineering the GP-CVaR Synthesis

A technical dissection of exact resolvent risk, Bayesian return shrinkage, and adaptive regime-switching algorithms.



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> INITIALIZING PORTFOLIO STUDIO ENV...

> LOAD: cuFolio [Brick Red]
> LOAD: MPDOK [Forest Green]
> LOAD: gp_cvar [Deep Cyan]
> LOAD: gp_cvar [Deep Cyan]
> LOAD: regime_switchers (smart1, smart2)
> STATUS: READY.
```

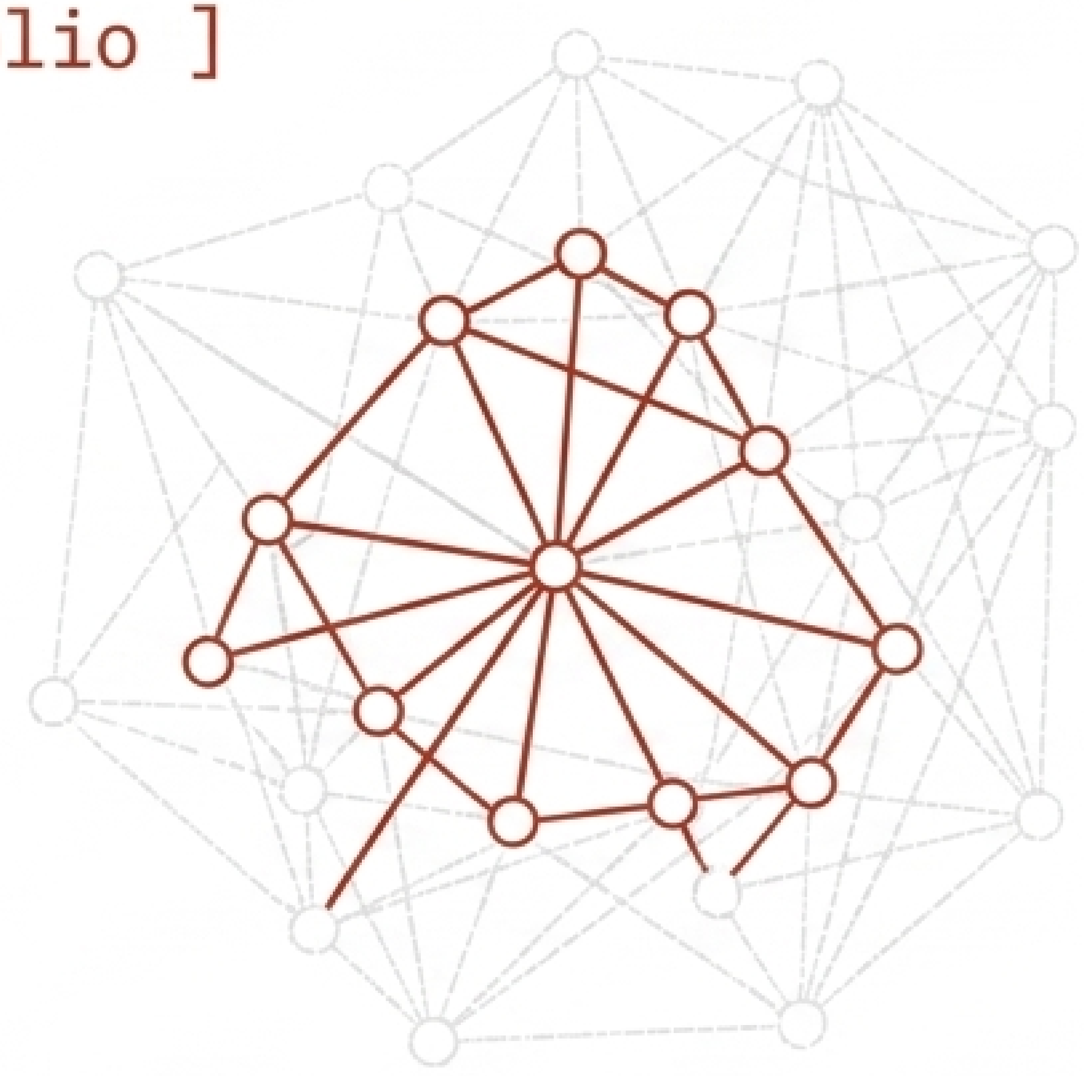
Three Axes of Optimization: The Strategy Matrix

Strategy	Return Input	Risk Objective	Risk Measure
Equal	n/a	n/a	1/N benchmark
cuFolio	raw historical mean	k=3 Neumann contagion overlay	variance
MPDOK	raw historical mean	exact resolvent $(I - \alpha\hat{A})^{-1}$	variance
gp_cvar	**CV-shrunk GP posterior mean**	**GP posterior covariance, scenario-sampled**	**CVaR95 (tail loss)**

Key Insight: cuFolio and MPDOK disagree on risk modeling depth but agree on the inputs. **gp_cvar** fundamentally rewires both the return estimate and the risk objective.

The Baseline Engines: Truncated vs. Exact Risk

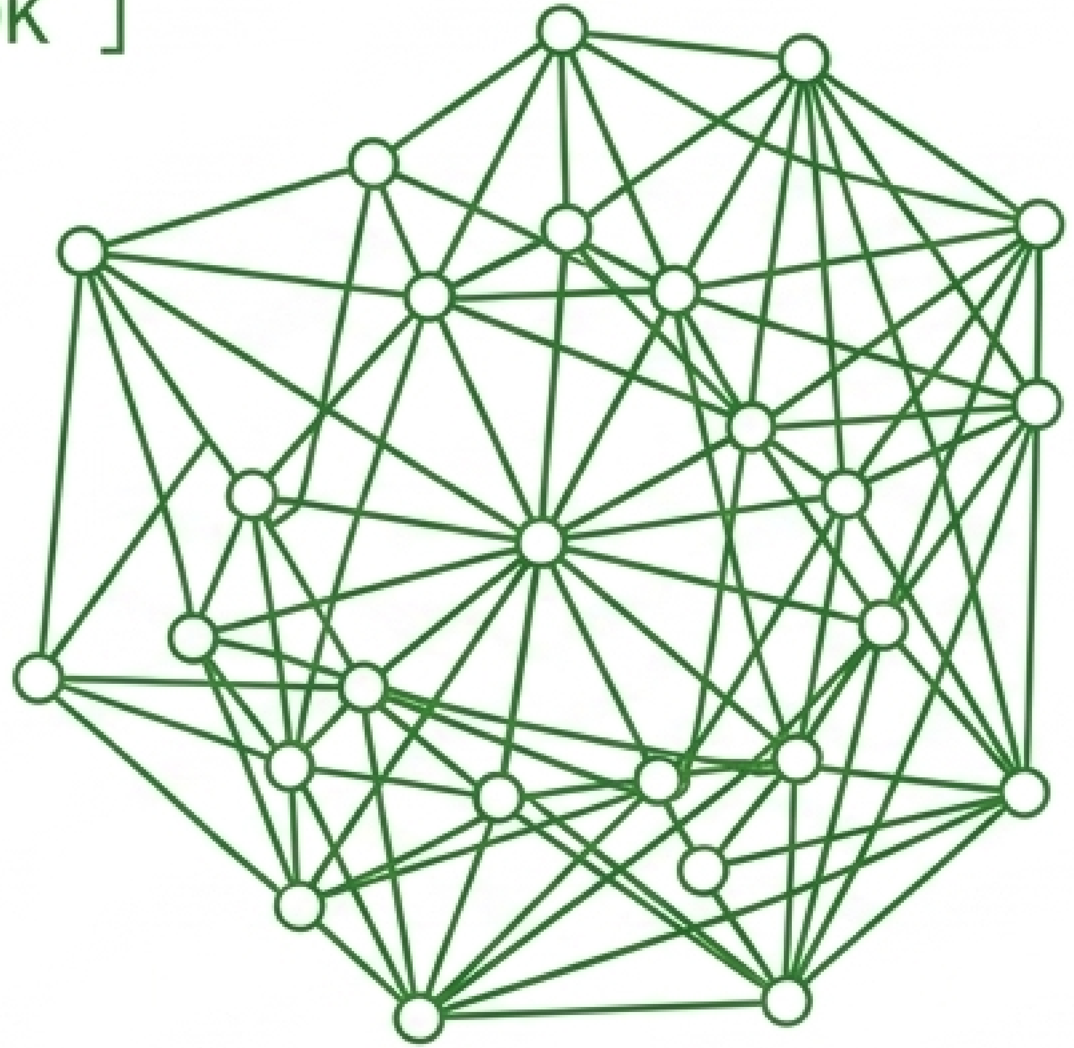
[cuFolio]



$$\tilde{\Sigma}_{ii} = \Sigma_{ii} \cdot (1 + \lambda \cdot sk^2)$$

Markowitz max-Sharpe. Cheap k=3 Neumann overlay underestimates densely connected cluster risk. Stays overweight in high-momentum stocks to capture bull market upside.

[MPDOK]

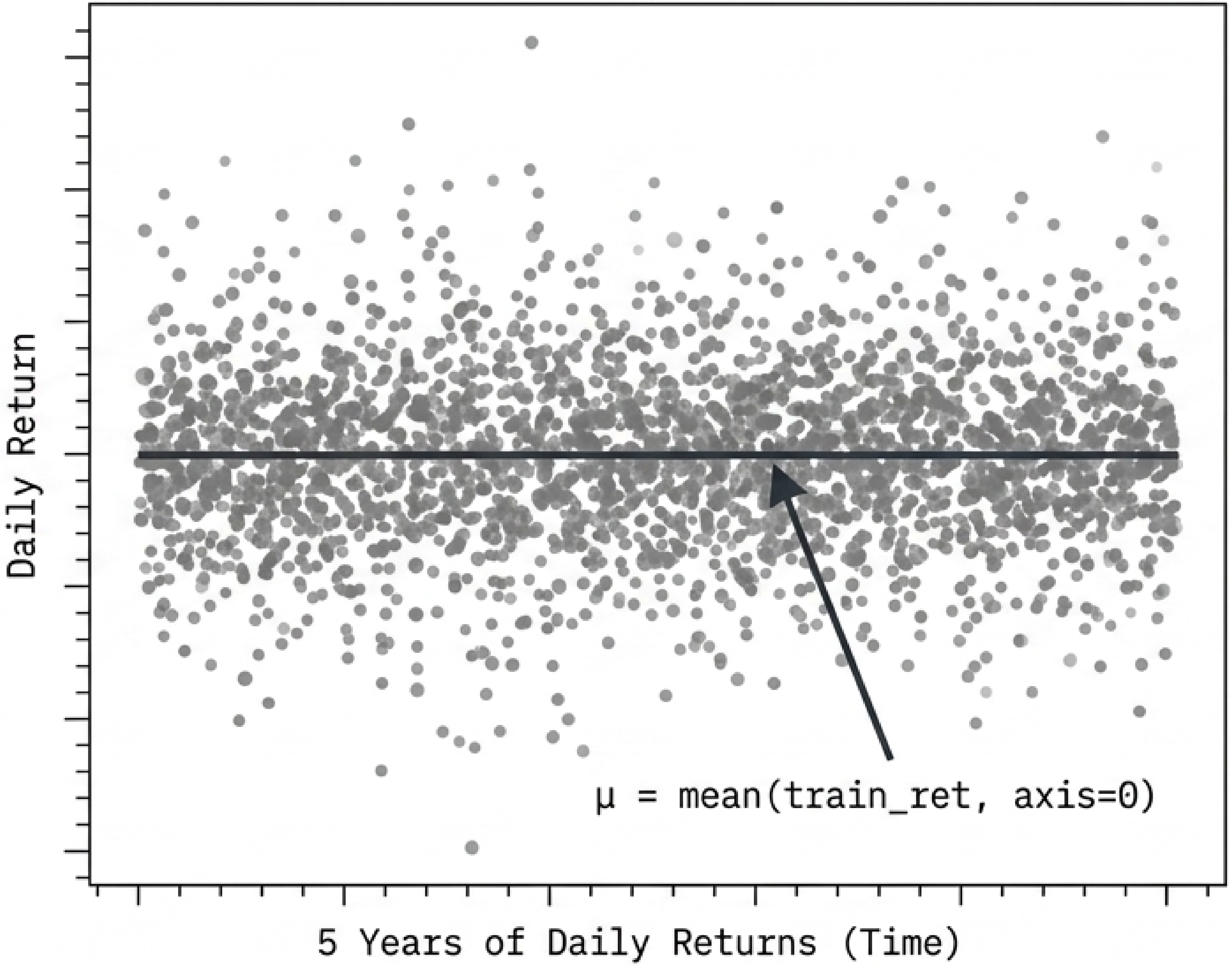


$$R = (I - \alpha\hat{A})^{-1} \quad | \quad \Sigma_m = \Sigma + \lambda R$$

Min-variance + systemic reach penalty. Hard caps any stock with exact reach > 0.70 at 0.5/N. Highly defensive in trending markets, superior in true contagion events.

The Shared Blindspot: Naive Point Estimates

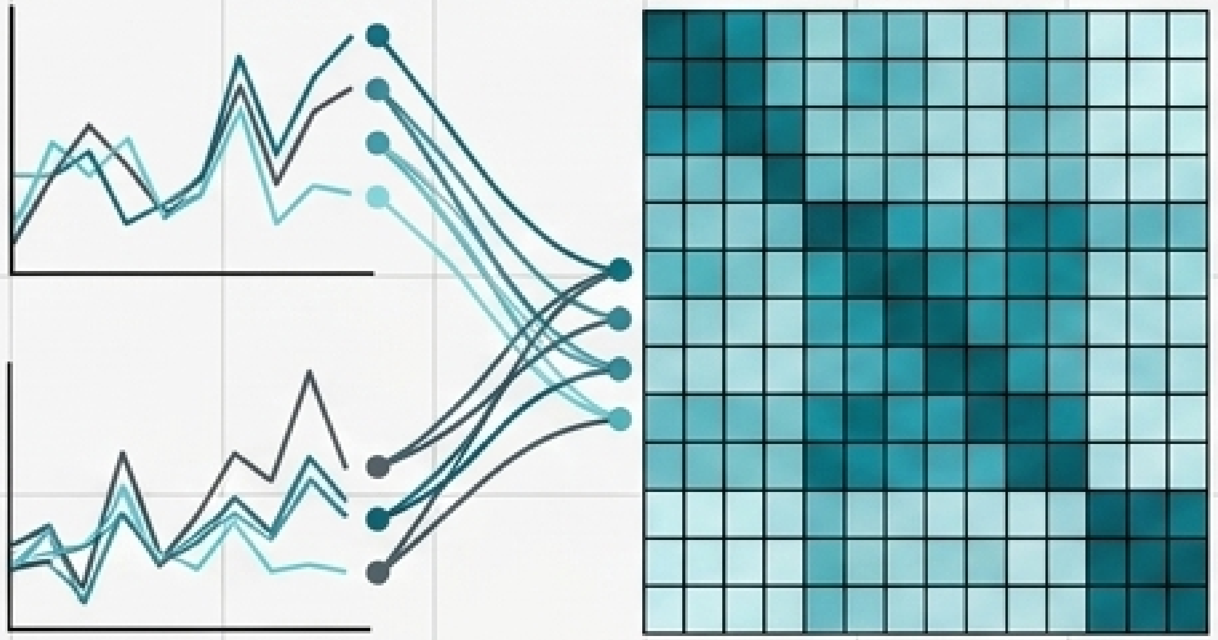
Both cuFolio and MPDOK take the raw historical sample mean return strictly at face value.



Optimization engines are notoriously hypersensitive to this noisy point estimate. Small changes in μ produce drastic, unstable changes in portfolio weights.

The gap: How do we extract a statistically honest expected return without breaking the studio's contagion network mechanics?

gp_cvar Stages 1 & 2: The Similarity Prior



[**Step 1:** The Asset-Similarity Kernel]

Uses a Matern-3/2 RKHS kernel to map asset similarity based on z-scored daily return histories.

[**Step 2:** GP Posterior solve via LUIRSolver]

$$K^{-1}(\mathbf{y} - \mathbf{m})$$

$\mathbf{m}$: constant mean offset

ℓ : kernel length-scale

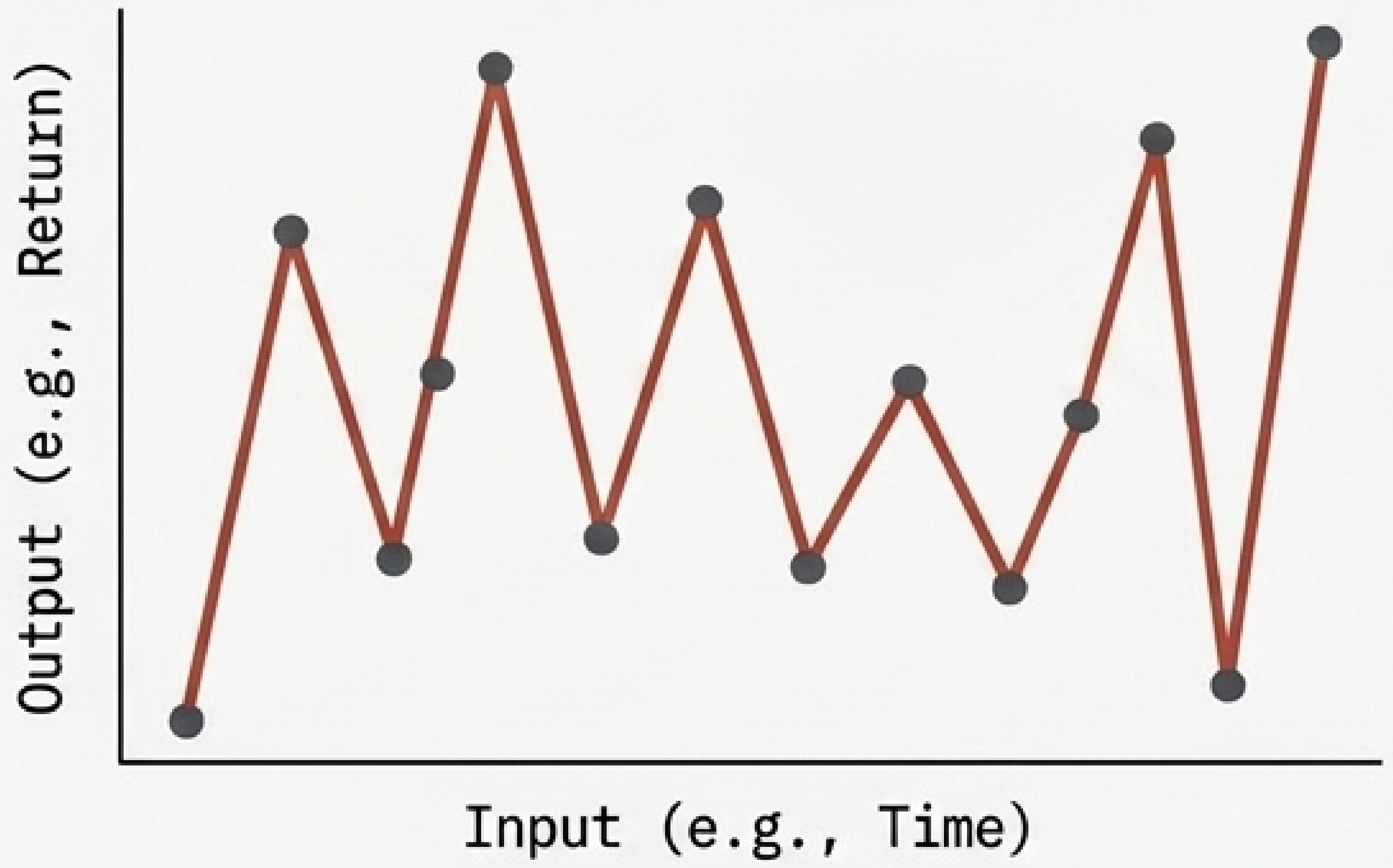
σ_f^2 : signal variance

σ_n^2 : noise variance

Cross-sectional guilt by association. An asset's expected return is pulled toward its kernel-neighbors. MPDOK's mixed-precision GMRES engine handles the dense linear solve via Nelder-Mead MLE.

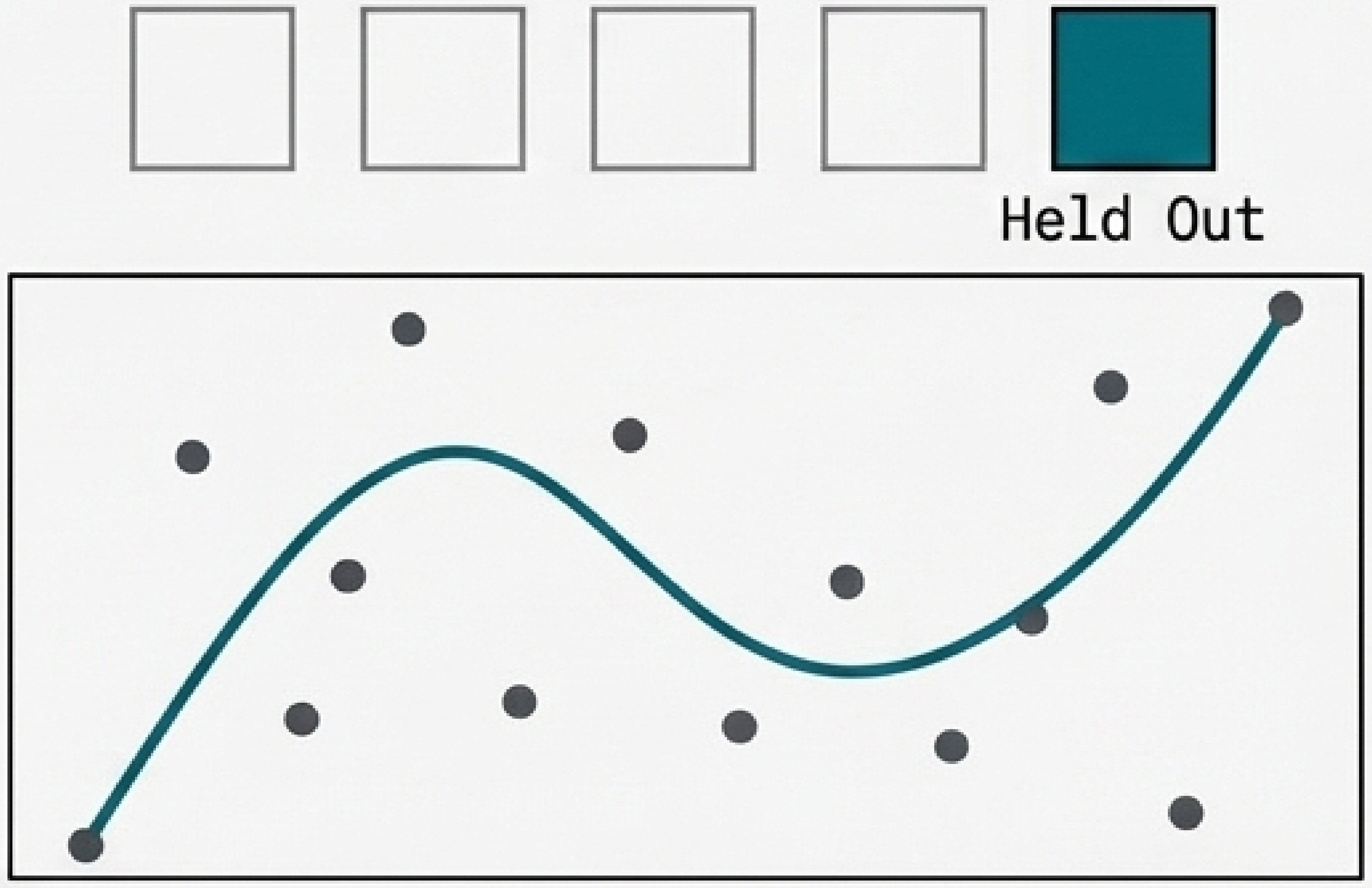
gp_cvar Stage 3: The In-Sample Trap & Out-of-Fold Shrinkage

Naive Posterior Fit (Overfitting)



As the Log Marginal Likelihood fit drives $\sigma_n^2 \to 0$, the mean interpolates targets exactly. It becomes circular, not Bayesian.

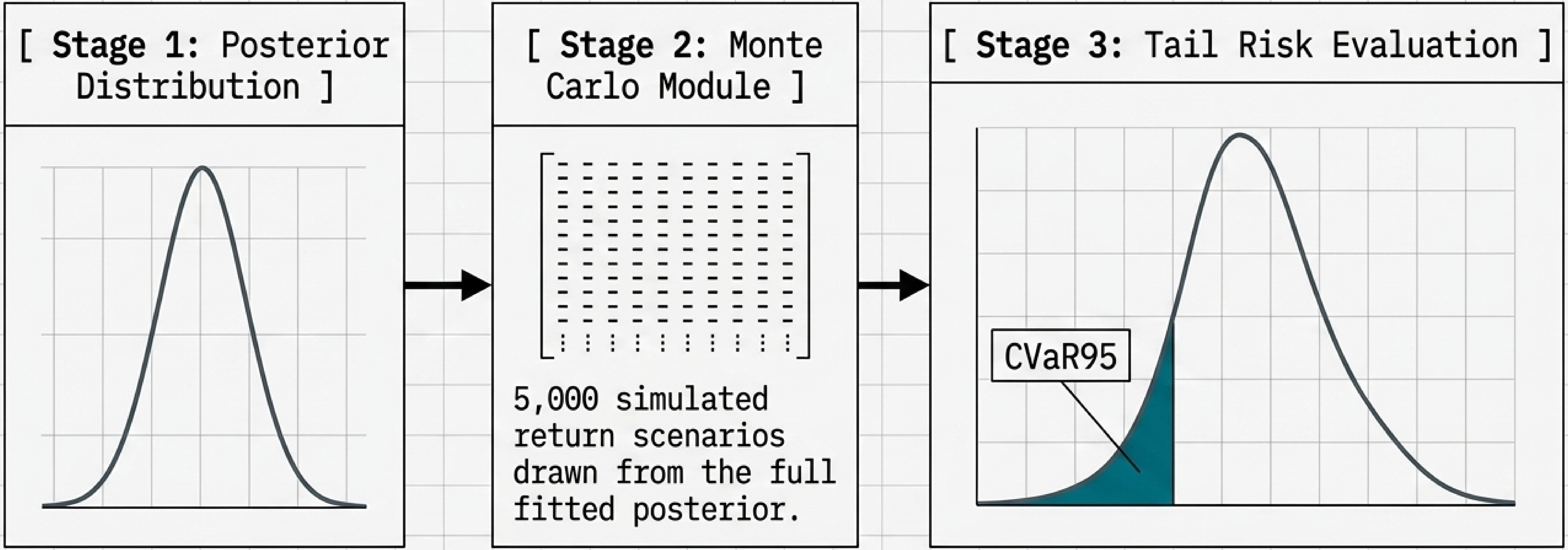
cv_shrunk_mean (Cross-Validation)



5-fold cross-validation. Each asset's shrunk-mean estimate comes from a GP fit that never saw that asset's own fold.

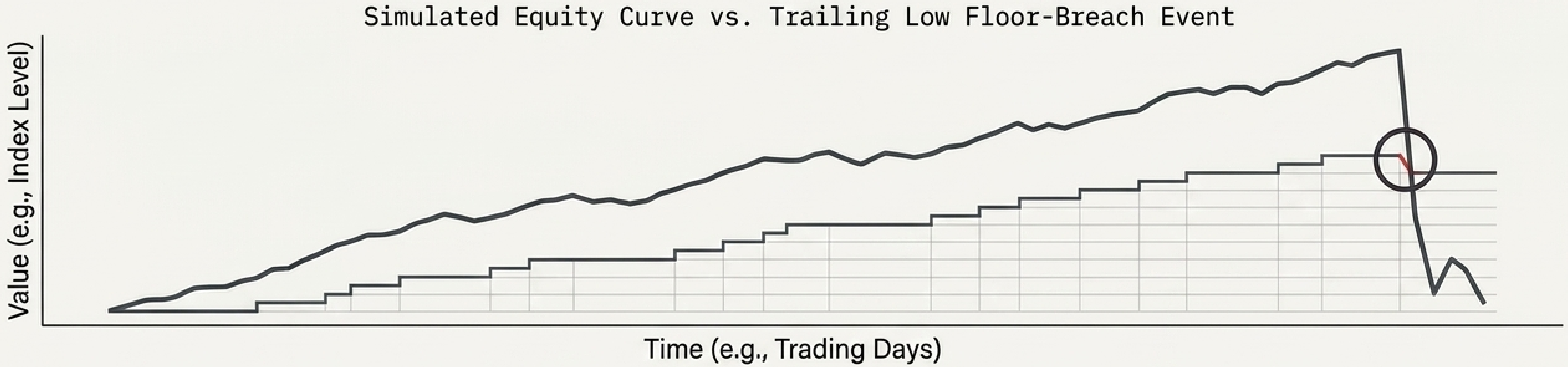
Result: Delivers ~29% lower RMSE against realized future-window returns vs naive grand-mean shrinkage.

gp_cvar Stage 4: Rockafellar-Uryasev Scenario LP



- **Shifting the objective:** Instead of penalizing symmetric variance (min $w'\Sigma w$), gp_cvar asks: “What is my expected loss in a bad 5% outcome?”
- Solves the classic Rockafellar-Uryasev Mean-CVaR95 scenario-LP directly from the GP posterior.

Regime Switching Logic: The Universal Floor-Breach Entry



[INPUT]
Equal-weight basket value.

[CONDITION 1]
Calculate trailing low over low_window (126 trading days).

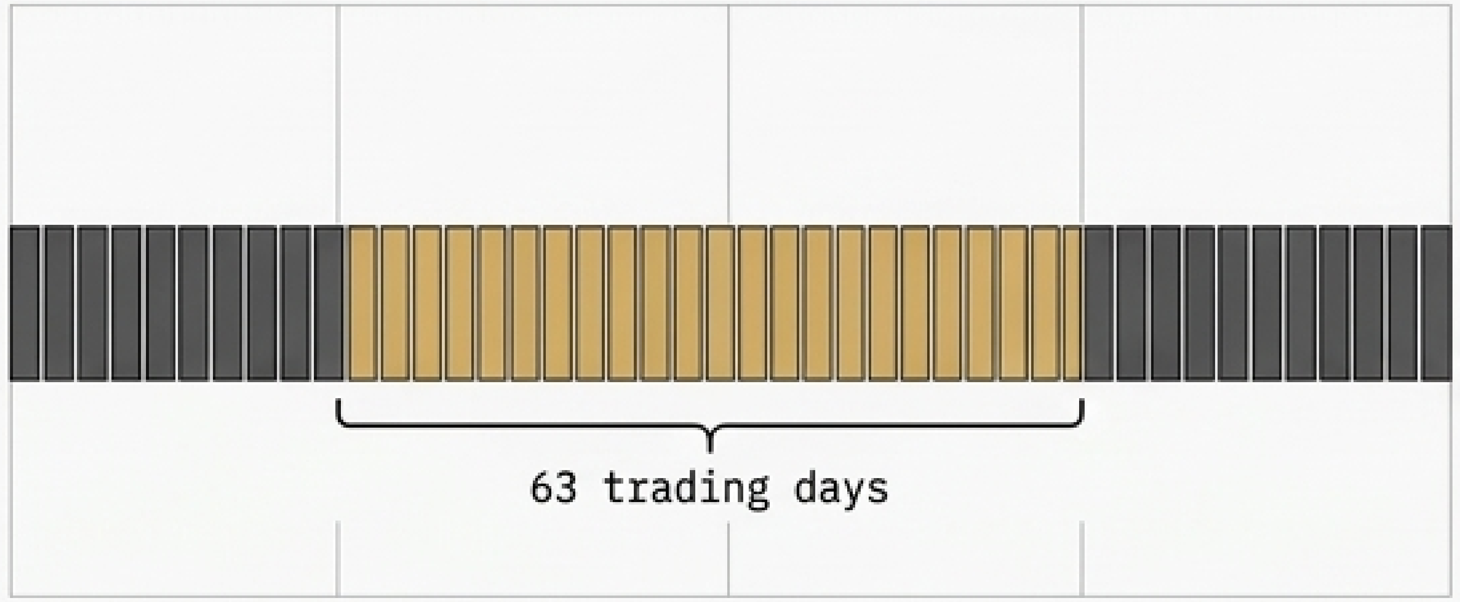
[CONDITION 2]
Current value < trailing low by breach_min (≥ 3%).

[ACTION]
Trigger RISK OFF. Enter MPDOK.

Key Insight: The floor rises with a bull market. Brief dips never touch it. Only a genuine structural reversal triggers th

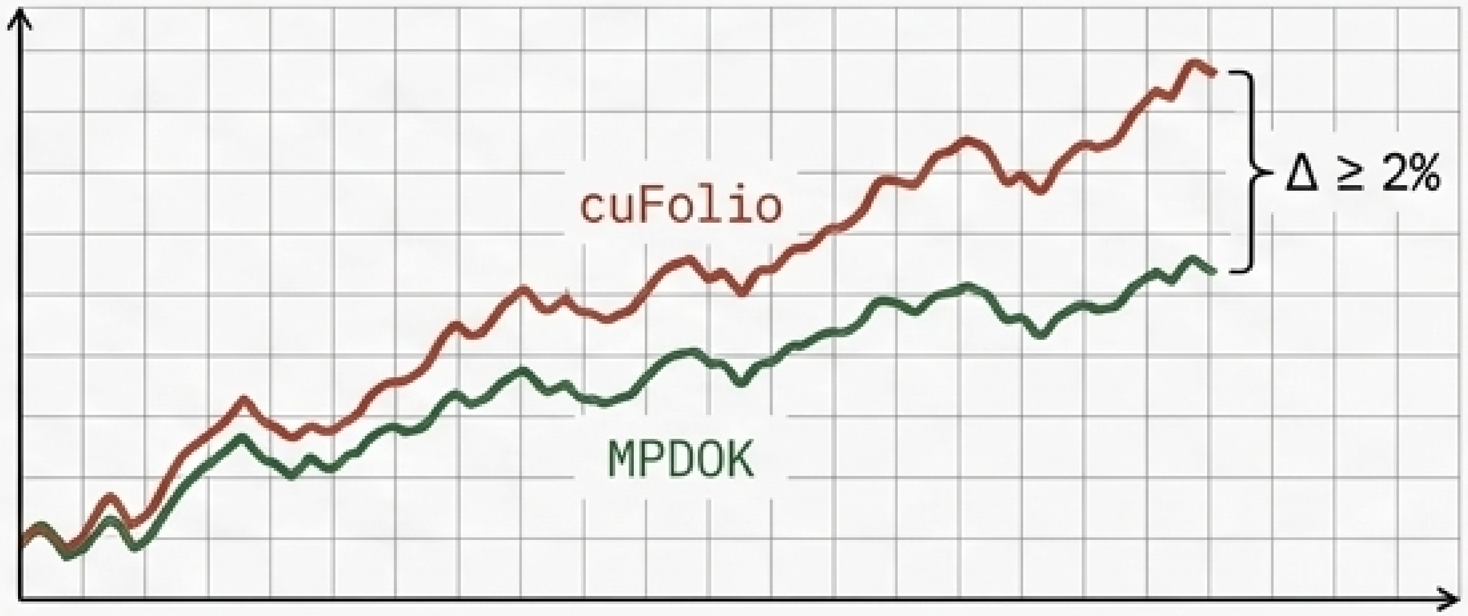
Smart1 Exit Sequence: Time-Gated Momentum Confirmation

Gate 1: Time Lock



min_mp_days. Must hold MPDOK for at least 63 trading days (~3 months).

Gate 2: Momentum Lock



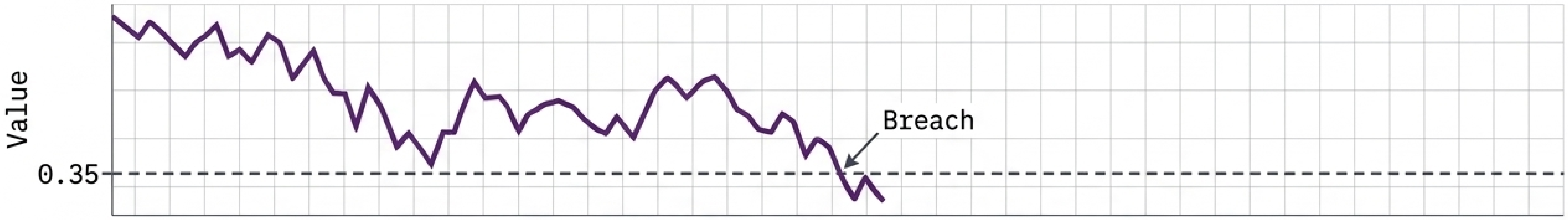
cuFolio's rolling return over those 63 days exceeds MPDOK by ≥ 2%.

Historical Context

In the fake Aug 2022 bear rally (S&P +17%), **Smart1** sat it out. The 63-day clock had not expired. It only switched to **RISK ON** (cuFolio) when genuine recovery was confirmed.

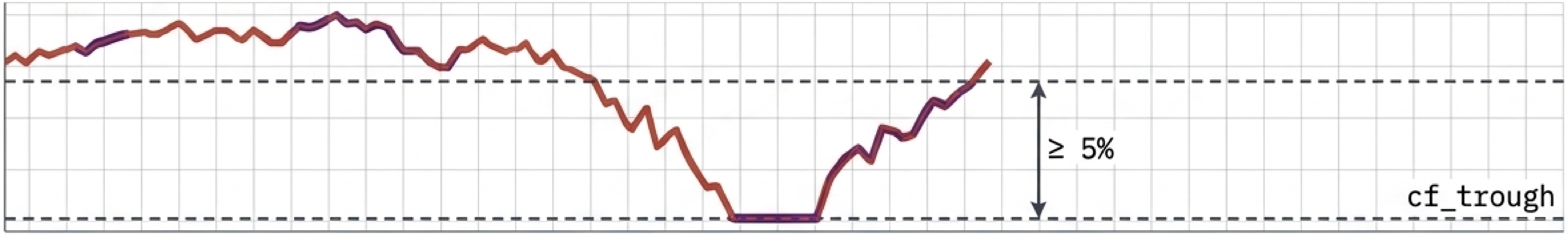
Smart2 Exit Sequence: Timeless Trough Anchoring

Gate 1: Network Destressed



The k=3 miss fraction (the gap) falls below 0.35. Instantaneous signal—no rolling averages, no lags.

Gate 2: Trough Bounce

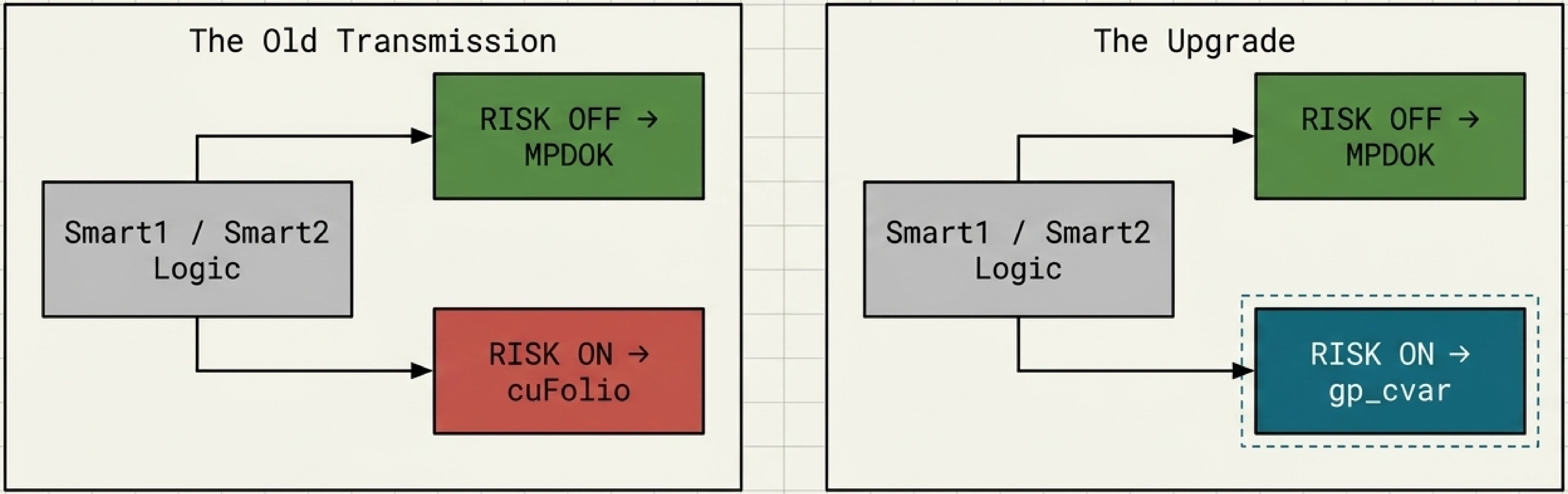


cuFolio bounces ≥ 5% above its running minimum since Smart2 entered MPDOK.

The Macro Contrast: Technical trough-bouncing beats the signed macro resolvent (TLT/Gold/VIX).
Equity markets are open systems; macro repricing lags. Smart2 captures bounces faster (e.g.,

Synthesis: The Upgraded Engine Swap (gp_smart)

Comparative Block Diagram: Architectural Upgrade



Core Thesis: Does the gp_cvar model's superior cross-validated return modeling and tail-risk optimization compound when wrapped in the studio's best-performing regime shells?

Phase 2 Backtest: Performance & Reality Checks

2020-2025 Test, 29 Assets				
Strategy	Ret	Vol	DD	SR
Equal	+204.1%	0.207	-36.4%	1.003
cuFolio	+323.2%	0.247	-30.4%	1.102
MPDOK	+131.6%	**0.170**	-29.1%	0.910
gp_cvar	**+330.5%**	0.242	**-29.4%**	**1.132**
smart2	**+336.3%**	0.222	-30.4%	1.221

READ WITH TWO HANDS

Regime Favorable: The backtest heavily favored a concentrated tech bet (NVDA/AAPL/MSFT/GOOGL/AMZN maxed at position caps). Only 9 of 29 assets active.

Covariance Proof: The RKHS covariance exhibits a 63% relative Frobenius distance from the sample correlation. Well-conditioned, but requires stress testing in a value-rotation regime before being declared settled truth.

The Ultimate Hybrid: Studio Architecture Summary

Data Layer
Raw Returns → cv_shrunk_mean (29% RMSE improvement) + RKHS Covariance.

↓

Engine Layer
GP Posterior → Rockafellar-Uryasev LP → gp_cvar

↓

Logic Layer
Floor Breach (3% drop over 126 days) → Switches to MPDOK

↓

Exit Layer
Gap < 0.35 AND Trough > 5% → Switches back to gp_cvar

Final Takeaway: By mathematically decoupling the return shrinkage (gp_cvar) from the exact-resolvent contagion mapping (MPDOK), and bridging them via technical trough-bouncing (gp_smart2), the studio creates an asymmetric structure: heavily defensive defensive during structural contagion, algorithmically aggressive during recovery.